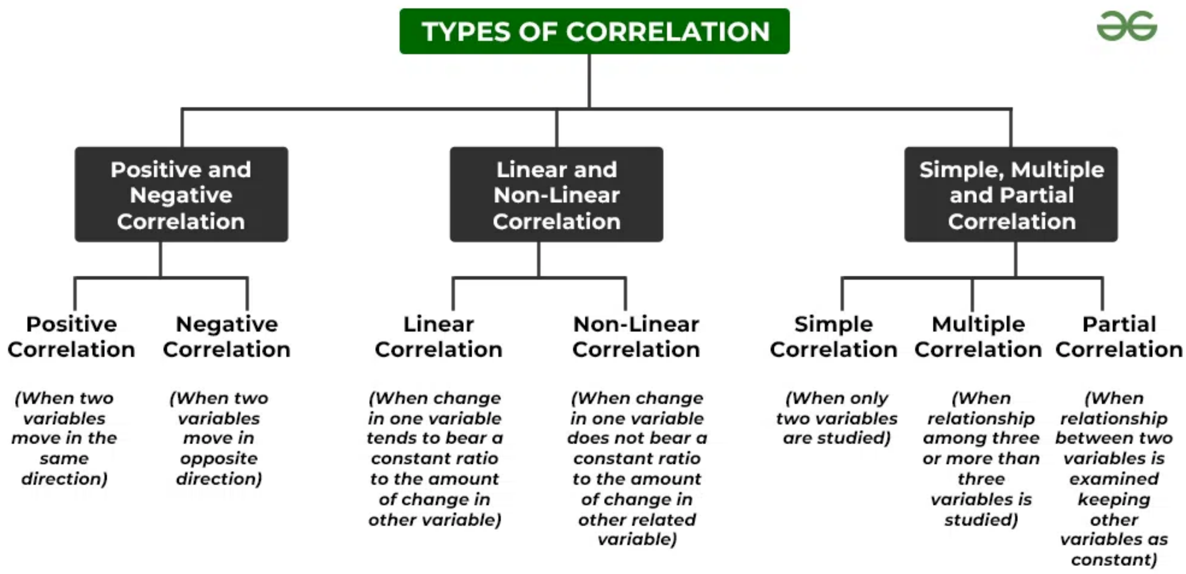
Correlation and Regression

Correlation:

What is Correlation?

Correlation analysis deals with the association between two or more variables. It is denoted by 'r'.

Types of Correlation



Degree of Correlation

The degree of correlation is measured through the coefficient of correlation. The degree of correlation for the given variables can be expressed in the following ways:

1. Perfect Correlation:

If the relationship between the two variables is in such a way that it varies in equal proportion (increase or decrease) it is said to be perfectly correlated. This can be of two types:

- **Positive Correlation:** When the proportional change in two variables is in the same direction, it is said to be positively correlated. In this case, the Coefficient of Correlation is shown as **+1**.

- **Negative Correlation:** When the proportional change in two variables is in the opposite direction, it is said to be negatively correlated. In this case, the Coefficient of Correlation is shown as **-1**.

2. Zero Correlation:

If there is no relation between two series or variables, it is said to have zero or no correlation. It means that if one variable changes and it does not have any impact on the other variable, then there is a lack of correlation between them. In such cases, the Coefficient of Correlation will be **0.**

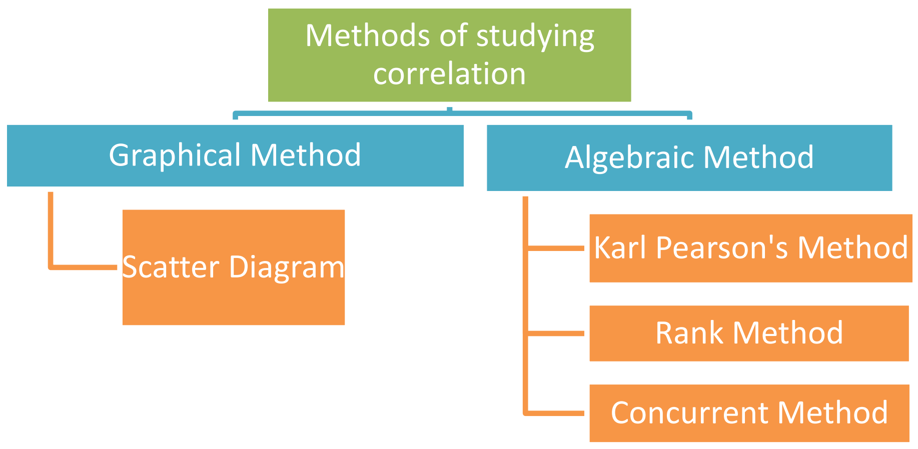
3. Low, High and Moderate Degree of Correlation

The degree of correlation can be **low** (when the coefficient of correlation lies between 0 and 0.25), **moderate** (when the coefficient of correlation lies between 0.25 and 0.75), or **high** (when the coefficient of correlation lies between 0.75 and 1).

Degree of Correlation

Degree of Correlation	Positive Correlation	Negative Correlation
Perfect Correlation	+1	-1
Very High Degree of Correlation	+0.9	-0.9
Fairly High Degree of Correlation	Between +0.75 and +0.9	Between -0.75 and -0.9
Moderate Degree of Correlation	Between +0.25 and +0.75	Between -0.25 and -0.75
Low Degree of Correlation	Between 0 and+0.25.	Between 0 and -0.25.
Zero/No Correlation (uncorrelated)	0	0

Methods of studying correlation:



1. **Karl Pearson's Method**
    a) Actual Mean Method
    b) Assumed Mean Method

a) Actual Mean Method:

The steps involved in the calculation of coefficient of correlation by using Actual Mean Method are:

1. Calculate the arithmetic mean of X series and Y series.
2. Calculate the deviation $(X - \bar{x})$ and $(Y - \bar{y})$, where $\bar{x}$ and $\bar{y}$ are mean of X and Y series respectively. Obtain the sum $\sum(X - \bar{x})$ and $\sum(Y - \bar{y})$.
3. Multiply the respective deviation and take the sum $\sum(X - \bar{x})(Y - \bar{y})$.
4. Calculate square of deviation, i.e, $(X - \bar{x})^2$ and $(Y - \bar{y})^2$ and find the sum $\sum(X - \bar{x})^2$ and $\sum(Y - \bar{y})^2$
5. Now, use the following formula to determine the Coefficient of Correlation:

$$\boldsymbol{r = \frac{\sum(X - \bar{x})(Y - \bar{y})}{\sqrt{\sum(X - \bar{x})^2}\sqrt{\sum(Y - \bar{y})^2}}}$$

Example:

Calculate Karl Pearson's coefficient of correlation by actual mean method:

Price (Rs)	10	12	14	16	18
Quatity (Units)	20	29	21	22	28

Solution

Let X denotes price and Y denotes quantity

X	$\bar{X} = 14$ $X - \bar{X}$	$(X - \bar{X})^2$	Y	$\bar{Y} = 24$ $Y - \bar{Y}$	$(Y - \bar{Y})^2$	$(X - \bar{X})(Y - \bar{Y})$
10	−4	16	20	−4	16	+16
12	−2	4	29	+5	25	−10
14	0	0	21	−3	9	0
16	+2	4	22	−2	4	−4
18	+4	16	28	+4	16	+16
$\sum X = 70$ $N = 5$	$\sum(X - \bar{X})$ $= 0$	$\sum(X - \bar{X})^2$ $= 40$	$\sum Y = 120$	$\sum(Y - \bar{Y})$ $= 0$	$\sum(Y - \bar{Y})^2$ $= 70$	$\sum(X - \bar{X})(Y - \bar{Y}$ $= 18$

$\bar{X} = \frac{\sum X}{N} = \frac{70}{5} = 14; \bar{Y} = \frac{\sum Y}{N} = \frac{120}{5} = 24$

$r = \frac{\sum(X-\bar{X})(Y-\bar{Y})}{\sqrt{\sum(X-\bar{X})^2}\sqrt{\sum(Y-\bar{Y})^2}} = \frac{18}{\sqrt{40}\sqrt{70}} = \frac{18}{\sqrt{2800}} = \frac{18}{52.91} = 0.34$

b) Assumed Mean Method

If your mean is in decimal, then we use assumed mean method

The steps involved in the calculation of coefficient of correlation by using Assumed Mean Method are:

1. Calculate the assumed mean for X series which will be the centre point of given data in X and similarly calculate assumed mean for Y series.
2. Take the deviation of X from assumed mean and denote it by d_x . Also calculate deviation of Y series from assumed mean and denote it by d_y.
3. Make a column and multiply d_x and d_y. Calculate square of d_x and d_y.
4. Calculate $\sum(d_x d_y)$, $\sum(d_x^2)$, $\sum(d_y^2)$, $\sum(d_x)$ and $\sum(d_y)$
5. Now, use the following formula to determine Coefficient of Correlation:

$$\boldsymbol{r} = \frac{\boldsymbol{N\sum d_x d_y - (\sum(d_x))(\sum((d_y))}}{\sqrt{\boldsymbol{N\sum d_x^2 - (\sum d_x)^2}} \times \sqrt{\boldsymbol{N\sum d_y^2 - (\sum d_y)^2}}}$$

Example:

Use Assumed Mean Method and determine the coefficient of correlation for the following data:

X	12	16	20	24	28	32	36
Y	6	9	12	15	18	21	24

Solution:

Calculation of Coefficient of Correlation (Assumed Mean Method)

X Series			**Y Series**			**dxdy**
X	**dx = X - A A = 20**	**dx^2**	**Y**	**dy = Y - A A = 12**	**dy^2**	
12	-8	64	6	-6	36	48
16	-4	16	9	-3	9	12
20 (A)	0	0	**12 (A)**	0	0	0
24	4	16	15	3	9	12
28	8	64	18	6	36	48
32	12	144	21	9	81	108
36	16	256	24	12	144	192
	$\sum dx = 28$	**$\sum dx^2 = 560$**		**$\sum dy = 21$**	**$\sum dy^2 = 315$**	**$\sum dxdy = 420$**

$$r = \frac{N\sum dxdy - \sum dx.\sum dy}{\sqrt{N\sum dx^2 - (\sum dx)^2}\sqrt{N\sum dy^2 - (\sum dy)^2}}$$

$$= \frac{(7\times420)-(28\times21)}{\sqrt{(7\times560)-(28)^2}\times\sqrt{(7\times315)-(21)^2}}$$

$$= \frac{2{,}940-588}{\sqrt{3{,}920-784}\times\sqrt{2{,}205-441}}$$

$$= \frac{2{,}352}{\sqrt{3{,}136}\times\sqrt{1{,}764}} = \frac{2{,}352}{56\times42}$$

$$= \frac{2{,}352}{2{,}352} = 1$$

Coefficient of Correlation = 1

It means that there is perfect positive correlation between the values of Series X and Series Y.

Karl Pearson's Coefficient of Correlation and Covariance

Karl Pearson's method of determining coefficient of correlation is based on the covariance of the given variables. **Covariance** is a statistical representation of the degree to which the two given variables vary together. Basically, Covariance is a number reflecting the degree to which the two variables vary together. The symbol of Covariance of two variables (say X and Y) is denoted by COV(X, Y).

$$\boldsymbol{COV(X,Y) = \frac{\sum(x-\overline{x})(y-\overline{y})}{N}}$$

The formula for calculating Karl Pearson's Coefficient of Correlation can be transformed into another easy formula as:

$$\boldsymbol{r = \frac{COV(X,Y)}{\sigma_x\sigma_y}}$$

Here, σ_x *and* σ_y are standard deviation of X and Y series respectively.

Example 1:

Determine the Coefficient of Correlation between X and Y.

	Series X	Series Y
Number of Items	30	30
Standard Deviation	4	3

The summation of the product of deviations of Series X and Y from their respective means is 200.

Solution:

Given: $N = 30, \ \ \sigma_x = 4, \ \ \sigma_y = 3 \ and \ COV(X,Y) = 200$

Apply the following formula,

$$r = \frac{COV(X,Y)}{\sigma_x \sigma_y} = \frac{200}{4 \times 3} = 0.5$$

Coefficient of Correlation = 0.5

It means that there is a positive correlation between X and Y.

Regression

What is Regression?

Regression analysis measures the nature and extent of the relation between two or more variables. Thus enables us to make prediction.

Types of Regression

For different types of Regression analysis, some assumptions need to be considered along with understanding the nature of variables and their distribution.

- Simple linear Regression
- Multi-linear Regression
- Polynomial Regression
- Time series Regression
- Logistic Regression

What is Simple-Linear Regression?

Simple-Linear Regression is a predictive model used for finding the ***linear*** relationship between a dependent variable and only one independent variables.

Here, 'Y' is our dependent variable, which is a continuous numerical and we are trying to understand how 'Y' changes with 'X'.

What is Multi-Linear Regression?

Multi-Linear Regression is a predictive model used for finding the ***linear*** relationship between a dependent variable and more than one independent variables.

Here, 'Y' is our dependent variable, which is a continuous numerical and we are trying to understand how 'Y' changes with independent variable 'X_1', 'X_2', 'X_3'… 'X_n' .

What is Logistic Regression?

When the dependent variable is discrete, the logistic regression technique is applicable. In other words, this technique is used to compute the probability of mutually exclusive occurrences such as pass/fail, true/false, 0/1, and so forth. Thus, the target variable can take on only one of two values, and a sigmoid curve represents its connection to the independent variable, and probability has a value between 0 and 1.

What is polynomial regression?

Polynomial Regression is a form of linear regression in which the relationship between the independent variable x and dependent variable y is modelled as an *nth-degree* polynomial.

The general form of a polynomial regression equation of degree n is:

$Y = a_0 + a_1x + a_2x^2 + \ldots + a_nx^n$

What is time series regression?

In case of time series regression, the data is dependent on time and the relation between the independent and dependent variables changes with time.

Regression Equation:

The regression equation is given as:

i) Equation of X on Y ==> X = a+bY
ii) Equation of Y on X ==> Y = a+bX

Methods for finding regression equation:

1. Regression equations using Normal equations
2. Regression equation using regression coefficient

1. Regression equations using Normal equations

Case I : Normal equation of X on Y

The equation of least square line is given by **X = a + bY**. We need to find a and b from given normal equation and substitute in the equation of least square line.

Normal equation:

a) $\sum X = Na + b\sum Y$
b) $\sum XY = a\sum Y + b\sum Y^2$

Case II : Normal equation of Y on X

The equation of least square line is given by **Y = a + bX**. We need to find a and b from given normal equation and substitute in the equation of least square line.

Normal equation:

a) $\sum Y = Na + b\sum X$
b) $\sum XY = a\sum X + b\sum X^2$

Question:

Consider the time series data given below:

x_i	8	3	2	10	11	3	6	5	6	8
y_i	4	12	1	12	9	4	9	6	1	14

Use the least square method to determine the equation of Y on X for the data.

Solution:

The equation of least square line is given by Y = a + bX

The Normal equations are:

a) $\sum Y = Na + b\sum X$
b) $\sum XY = a\sum X + b\sum X^2$

X	Y	X^2	XY
8	4	64	32
3	12	9	36
2	1	4	2
10	12	100	120
11	9	121	99
3	4	9	12
6	9	36	54
5	6	25	30
6	1	36	6
8	14	64	112
$\sum X = 62$	$\sum Y = 72$	$\sum X^2 = 468$	$\sum XY = 503$

Substituting these values in the normal equations,

10a + 62b = 72 ====> (eq1)

62a + 468b = 503 ====> (eq2)

Multiplying (eq1) with 62 and (eq2) with 10 and take the difference,

620a + 3844b – (620a + 4680b) = 4464 – 5030

-836b = -566

b = 566/836

b = 283/418

b = 0.677

Substituting b = 0.677 in equation (1),

10a + 62(0.677) = 72

10a + 41.974 = 72

10a = 72 – 41.974

10a = 30.026

a = 30.026/10

a = 3.0026

Therefore, the equation becomes,

Y = a + bX

y = 3.0026 + 0.677x

Question:

Consider the time series data given below:

X	1	2	3	4	5
Y	2	5	3	8	7

Use the least square method to determine the equation of X on Y for the data.

Solution:

The equation of least square line is given by X = a + bY

The Normal equations are:

a) $\sum X = Na + b\sum Y$
b) $\sum XY = a\sum Y + b\sum Y^2$

X	Y	Y^2	XY
1	2	4	2
2	5	25	10
3	3	9	9
4	8	64	32
5	7	49	35
$\sum X = 15$	$\sum Y = 25$	$\sum Y^2 = 151$	$\sum XY = 88$

Substituting these values in the normal equations,

5a + 25b = 15 ====> (eq1)

25a + 121b = 88 ====> (eq2)

On solving this equation we get,

a= 0.5 and b=0.5

Therefore, the equation becomes,

X = a + bY

X = 0.5 + 0.5b

2. Regression Equation using Regression Coefficient Method

a) Using actual values of X and Y series
b) Using deviation from actual mean
c) Using deviations from Assumed mean

a) Regression equation using actual values of X and Y series

Case I: X on Y

$$\textbf{Eq:} \qquad \boldsymbol{X - \bar{X} = b_{xy}(Y - \bar{Y})}$$

Where b_{xy} is regression coefficient given as

$$\boldsymbol{b_{xy} = \frac{N\sum XY - \sum X \sum Y}{N\sum Y^2 - (\sum Y)^2}}$$

$\bar{X}$ *and* $\bar{Y}$ is mean of X and Y series respectively.

Case II: Y on X

$$\textbf{Eq: } \boldsymbol{Y - \bar{Y} = b_{yx}(X - \bar{X})}$$

Where b_{yx} is regression coefficient given as

$$\boldsymbol{b_{yx} = \frac{N\sum XY - \sum X \sum Y}{N\sum X^2 - (\sum X)^2}}$$

$\bar{X}$ *and* $\bar{Y}$ is mean of X and Y series respectively.

Example:

Calculate the regression equations of X on Y from the following data:

x	1	2	3	4	5
y	2	5	3	8	7

Case I: X on Y

The given equation is $X - \bar{X} = b_{xy}(Y - \bar{Y})$

Therefore, we need to calculate $b_{xy} = \frac{N\sum XY - \sum X \sum Y}{N\sum Y^2 - (\sum Y)^2}$

X	Y	XY	Y^2
1	2	2	4
2	5	10	25
3	3	9	9
4	8	32	64
5	7	35	49
$\sum X = 15$	$\sum Y = 25$	$\sum XY = 88$	$\sum Y^2 = 151$

$\bar{X} = \frac{\sum X}{N} = \frac{15}{5} = 3$ and $\bar{Y} = \frac{\sum Y}{N} = \frac{25}{5} = 5.$

Now, substitute all the values obtained to find b_{xy}.

Therefore, $b_{xy} = \frac{N\sum XY - \sum X \sum Y}{N\sum Y^2 - (\sum Y)^2} = \frac{5 \times 88 - 15 \times 25}{5 \times 151 - (25)^2} = \frac{440 - 375}{755 - 625} = \frac{65}{130} = 0.5$

Now, find the regression equation given as $X - \bar{X} = b_{xy}(Y - \bar{Y})$

$$=> \; X - 3 = 0.5(Y - 5)$$

On simplifying further, we get

$$X = 0.5 + 0.5Y$$

b) Regression equation using deviation from actual mean.

Case I: X on Y

$$\textbf{Eq:} \quad \boldsymbol{X - \bar{X} = b_{xy}(Y - \bar{Y})}$$

Where b_{xy} is regression coefficient given as

$$\boldsymbol{b_{xy} = \frac{\sum (X - \bar{X})(Y - \bar{Y})}{\sum (Y - \bar{Y})^2}}$$

$\bar{X}$ *and* $\bar{Y}$ is mean of X and Y series respectively.

Case II: Y on X

$$\textbf{Eq:} \quad \boldsymbol{Y - \bar{Y} = b_{yx}(X - \bar{X})}$$

Where b_{yx} is regression coefficient given as

$$\boldsymbol{b_{yx} = \frac{\sum(X - \bar{X})(Y - \bar{Y})}{\sum(X - \bar{X})^2}}$$

$\bar{X}$ *and* $\bar{Y}$ is mean of X and Y series respectively.

Example: Obtain regression equation Y on X from the following data

X	2	4	6	8	10	12
Y	4	2	5	10	3	6

Solution:

Given equation $Y - \bar{Y} = b_{yx}(X - \bar{X})$

We need to calculate $b_{yx} = \frac{\sum(X-\bar{X})(Y-\bar{Y})}{\sum(X-\bar{X})^2}$

$\bar{X} = \frac{\sum X}{N} = \frac{42}{6} = 7$ and $\bar{Y} = \frac{\sum Y}{N} = \frac{30}{6} = 5$

X	Y	$X - \bar{X}$	$Y - \bar{Y}$	$(X - \bar{X})(Y - \bar{Y})$	$(X - \bar{X})^2$
2	4	-5	-1	5	25
4	2	-3	-3	9	9
6	5	-1	0	0	1
8	10	1	5	5	1
10	3	3	-2	-6	9
12	6	5	1	5	25
$\sum X = 42$	$\sum Y = 30$			$\sum(X - \bar{X})(Y - \bar{Y}) = 18$	$\sum(X - \bar{X})^2 = 70$

Therefore, $b_{yx} = \frac{\sum(X-\bar{X})(Y-\bar{Y})}{\sum(X-\bar{X})^2} = \frac{18}{70} = 0.25$

Now, find the regression equation given as $Y - \bar{Y} = b_{yx}(X - \bar{X})$

$$=> Y - 5 = 0.25(X - 7)$$

On simplifying we get,

$$Y = 3.2 + 0.25X$$

Regression equation from coefficient of correlation

Case I: X on Y

$$\textbf{Eq: } \boldsymbol{X - \bar{X} = b_{xy}(Y - \bar{Y})}$$

Where b_{xy} is regression coefficient given as

$$\boldsymbol{b_{xy} = \frac{r\sigma_x}{\sigma_y}}$$

$\bar{X}$ *and* $\bar{Y}$ is mean of X and Y series respectively. σ_x is standard deviation for X and σ_y is standard deviation for Y. r is correlation coefficient.

Case II: Y on X

$$\textbf{Eq: } \quad \boldsymbol{Y - \bar{Y} = b_{yx}(X - \bar{X})}$$

Where b_{yx} is regression coefficient given as

$$\boldsymbol{b_{yx} = \frac{r\sigma_y}{\sigma_x}}$$

$\bar{X}$ *and* $\bar{Y}$ is mean of X and Y series respectively. σ_x is standard deviation for X and σ_y is standard deviation for Y. r is correlation coefficient.

Example:

You are given the following information

i) Obtain two regression equations
ii) Estimate Y when X=9
iii) Estimate X when Y =12

	X	Y
Arithmetic Mean	5	12
Standard deviation	2.6	3.6
Correlation Coefficient	r=0.7	

i) Obtain two regression equations

Case I: X on Y

Eq: $X - \bar{X} = b_{xy}(Y - \bar{Y})$

$X - 5 = b_{xy}(Y - 12)$. Now calculate b_{xy}

$$b_{xy} = \frac{0.7 \times 2.6}{3.6} = 0.50$$

Therefore, $X - 5 = 0.50 \times (Y - 12)$

⇨ X = 0.50Y – 1 -------------> 1^{st} Eq

Case II: Y on X

Eq: $Y - \bar{Y} = b_{yx}(X - \bar{X})$

: $Y - 12 = b_{yx}(X - 5)$. Now calculate b_{xy}

$$b_{yx} = \frac{0.7 \times 3.6}{2.6} = 0.96$$

Therefore, $Y - 12 = 0.96 \times (X - 5)$

Y = 0.96X + 7.2 -----------------> 2^{nd} Eq

ii) Estimate Y when X=9

For estimating (finding) value of Y when X is given, use equation case Y on X

Consider Eq 2^{nd}

Y = 0.96X +7.2

Y = 0.96×9 +7.2

Y = 8.64 +7.2

Y= 15.84.

iii) Estimate X when Y =12

For estimating (finding) value of Y when X is given, use equation case Y on X

Consider Eq 1^{st}

X = 0.50Y – 1

X= 0.50×12 - 1

X= 6 – 1

X=5